CHAPTER 2
Getting to the Shell

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In the old days of Linux, system administrators, programmers, and system users all sat at something called a Linux console terminal entering shell commands and viewing text output. These days, with graphical desktop environments, it's getting harder to find a shell prompt on the system in order to enter shell commands. This chapter discusses what is required to reach a command-line environment. And it walks you through a few terminal emulation packages you may run into in the various Linux distributions.

Reaching the Command Line

Before the days of graphical desktops, the only way to interact with a Unix system was through a text *command-line interface (CLI)* provided by the shell. The CLI allowed text input only and could display only text and rudimentary graphics output.

Because of these restrictions, output devices were not very fancy. Often, you needed only a simple dumb terminal to interact with the Unix system. A dumb terminal was usually nothing more than a monitor and keyboard connected to the Unix system via a communication cable (usually a multiwire serial cable). This simple combination provided an easy way to enter text data into the Unix system and view text results.

As you well know, things are significantly different in today's Linux environment. Just about every Linux desktop distribution uses some type of graphical desktop environment. However, to enter shell commands, you still need a text display to access the shell's CLI. The problem now is getting to one. Sometimes finding a way to get a CLI in a Linux distribution is not an easy task.

Console terminals

One way to get to a CLI is to access the Linux system via text mode. This provides nothing more than a simple shell CLI on the monitor, just like the days before graphical desktops. This mode is called the *Linux console* because it emulates the old days of a hard-wired console terminal and is a direct interface to the Linux system.

When the Linux system starts, it automatically creates several *virtual consoles*. A virtual console is a terminal session that runs in Linux system memory. Instead of having several dumb terminals connected to the computer, most Linux distributions start five or six (or sometimes even more) virtual consoles that you can access from a single computer keyboard and monitor.

Graphical terminals

The alternative to using a virtual console terminal is to use a *terminal emulation package* from within the Linux graphical desktop environment. A terminal emulation package simulates working on a console terminal but within a desktop graphical window. Figure 2-1 shows an example of a terminal emulator running in a Linux graphical desktop environment.

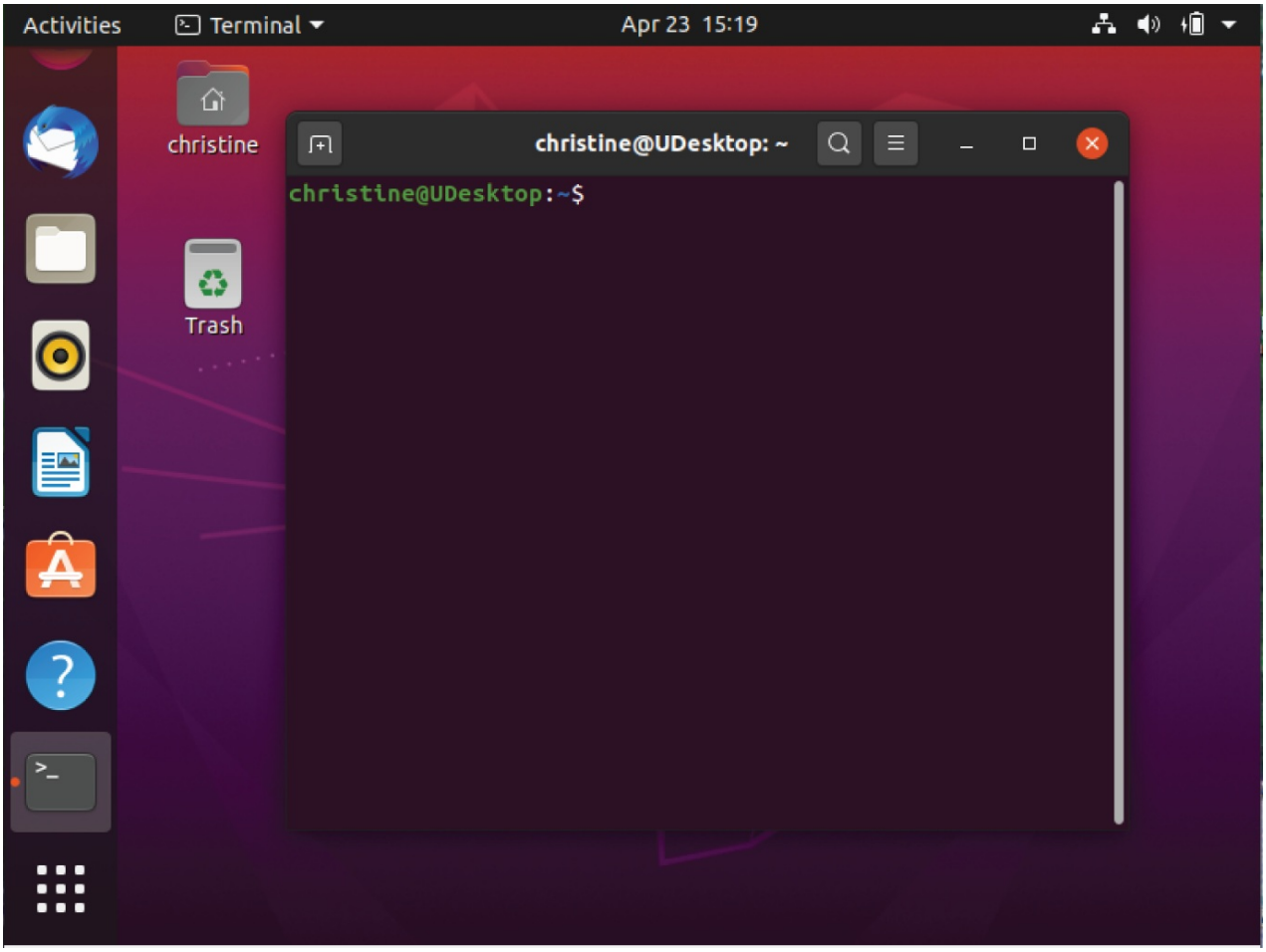

FIGURE 2-1 A simple terminal emulator running on a Linux desktop

Graphical terminal emulation is responsible only for a portion of the Linux graphical experience. As a whole, the experience is accomplished through several components, including graphical terminal emulation software (called a *client*). Table 2-1 shows the different components in the Linux graphical desktop environment.

TABLE 2-1 Graphical Interface Elements

Name	Examples	Description
Client	Graphical terminal emulator, desktop environment (GNOME Shell, KDE Plasma), network browser	An application that requests graphical services
Display Server	Wayland, X Window System	Element that manages the display (screen) and the input devices (keyboard, mouse, touch screen)
Window Manager	Mutter, Metacity, Kwin	Element that adds borders to windows and provides features to move and manage windows
Widgets Library	Plasmoids, Cinnamon Spices	Element that adds menus and appearance items for desktop environment clients

For dealing with the command line from the desktop, the focus is on the graphical terminal emulator. You can think of graphical terminal emulators as CLI terminals *in* the graphical user interface (GUI) and virtual console terminals as CLI terminals *outside* the GUI. Understanding the various terminals and their features can enhance your command-line experience.

Accessing CLI via a Linux Console Terminal

In the early days of Linux, when you booted up your system you would see a login prompt on your monitor, and that's all. As mentioned earlier, this is called the Linux console. It was the only place you could enter commands for the system.

Even though several virtual consoles are created at boot time, many Linux desktop distributions switch to a graphical environment after the boot sequence completes. This provides the user with a graphical login and desktop experience. For these systems, accessing a virtual console is done manually.

On most Linux distributions, you can access one of the Linux virtual consoles using a simple keystroke combination. Usually, you must hold down the Ctrl+Alt key combination and then press a function key (F1 through F7) for the virtual console you want to use. Function key F2 produces virtual console 2, key F3 produces virtual console 3, key F4 produces virtual console 4, and so on.

NOTE

Linux distributions typically use the Ctrl+Alt key combination with F1, F7, or F8 to reach the graphical interface. Ubuntu and CentOS both use F1. However, it is best to test and see where your distribution puts the graphical interface, especially on older distribution versions.

Text mode virtual consoles use the whole screen and start with the text login screen displayed. An example of a text login screen from a virtual console is shown in Figure 2-2.

Notice in Figure 2-2 the word `tty2` at the end of the first text line. The `2` in `tty2` indicates that it is virtual console 2 and was reached by pressing the Ctrl+Alt+F2 key sequence. `tty` stands for teletypewriter. *Teletypewriter* is an old term, indicating a machine used for sending messages.

```
Ubuntu 20.04 LTS UDesktop tty2

UDesktop login: christine
Password:
Welcome to Ubuntu 20.04 LTS (GNU/Linux 5.4.0-26-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage


0 updates can be installed immediately.
0 of these updates are security updates.

Your Hardware Enablement Stack (HWE) is supported until April 2025.
Last login: Fri Apr 24 17:02:52 EDT 2020 on tty2
christine@UDesktop:~$ _
```

FIGURE 2-2 Linux virtual console login screen

NOTE

Not all Linux distributions show the virtual console's `tty` number at the login screen. If you are logged into a virtual console, you can enter the command tty and press Enter to see what virtual console you are currently using. Entering commands is covered in Chapter 3, "Basic Bash Shell Commands."

You log into a console terminal by entering your user ID after the `login:` prompt and typing your password after the `Password:` prompt. If you have never logged in this way before, be aware that typing your password is a different experience than it is in a graphical environment. In a graphical environment, you may see dots or asterisks indicating the password characters as you type. However, at the virtual console, *nothing* is displayed when you type your password.

NOTE

Keep in mind that, within the Linux virtual console, you do not have the ability to run any graphical programs.

After logging into a virtual console, you are taken to the Linux CLI, and you can switch to another virtual console

without losing your current active session. You can switch between all the virtual consoles, with multiple active sessions running. This feature provides a great deal of flexibility while you work at the CLI.

Additional flexibility deals with the virtual console's appearance. Even though it is a text mode console terminal, you can modify the text and background colors.

For example, it may be easier on your eyes to set the background of the terminal to white and the text to black. After you have logged in, you can accomplish this modification in a couple of ways. One way is to type in the command **`setterm --inversescreen on`** and press the Enter key, as shown in Figure 2-3. Notice in the figure that the `--inversescreen` feature is being turned on using the option `on` . You can also turn it off using the `off` option.

```
CentOS Linux 8 (Core)
Kernel 4.18.0-147.5.1.el8_1.x86_64 on an x86_64

Activate the web console with: systemctl enable --now cockpit.socket

localhost login: christine
Password:
Last login: Sat Apr 25 11:30:55 on tty3
[christine@localhost ~]$
[christine@localhost ~]$ tty
/dev/tty3
[christine@localhost ~]$ setterm --inversescreen on
[christine@localhost ~]$
```

FIGURE 2-3 Linux virtual console with `inversescreen` being turned on

Another way is to type two commands, one after the other. Type **`setterm --background white`** and press Enter, and then type **`setterm --foreground black`** and press Enter. Be careful because, when you change your terminal background first, it may be hard to see the commands you are typing.

With the commands in the preceding paragraph, you are not turning features on and off, as with `--inversescreen` . Instead, you have a choice of eight colors. The choices are `black` , `red` , `green` , `yellow` , `blue` , `magenta` , `cyan` , and `white` (which looks gray on some distributions). You can get rather creative with your plain text mode console terminals. Table 2-2 shows some options you can use with the `setterm` command to help improve your console terminal's readability or appearance.

TABLE 2-2 `setterm` **Options for Foreground and Background Appearance**

Option	Parameter Choices	Description
`--background`	`black` , `red` , `green` , `yellow` , `blue` , `magenta` , `cyan` , or `white`	Changes the terminal's background color to the one specified
`--foreground`	`black` , `red` , `green` , `yellow` , `blue` , `magenta` , `cyan` , or `white`	Changes the terminal's foreground color, specifically text, to the one specified
`--inversescreen`	`on` or `off`	Switches the background color to the foreground color and the foreground color to the background color
`--reset`	None	Changes the terminal appearance back to its default setting and clears the screen
`--store`	None	Sets the current terminal's foreground and background colors as the values to be used for `--reset`

Virtual console terminals are great for accessing the CLI outside the GUI. However, sometimes, you need to access the CLI and run graphical programs. Using a terminal emulation package solves this problem and is a popular way to access the shell CLI from within the GUI. The following sections describe common software packages that provide graphical terminal emulation.

Accessing CLI via Graphical Terminal Emulation

The graphical desktop environment offers a great deal more variety for CLI access than the virtual console terminal does. Many terminal emulator applications are available for the graphical environment. Each package provides its

own unique set of features and options. Some popular graphical terminal emulator applications are shown in Table 2-3 along with their websites.

TABLE 2-3 Popular Graphical Terminal Emulator Packages

Name	Website
Alacritty	github.com/alacritty/alacritty
cool-retro-term	github.com/Swordfish90/cool-retro-term
GNOME Terminal	wiki.gnome.org/Apps/Terminal
Guake	guake-project.org
Konsole	konsole.kde.org
kitty	sw.kovidgoyal.net/kitty
rxvt-unicode	software.schmorp.de/pkg/rxvt-unicode.html
Sakura	pleyades.net/david/projects/sakura
st	st.suckless.org
Terminator	gnometerminator.blogspot.com
Terminology	enlightenment.org/about-terminology.md
Termite	github.com/thestinger/termite
Tilda	github.com/lanoxx/tilda
xterm	invisible-island.net/xterm
Xfce4-terminal	docs.xfce.org/apps/terminal/start
Yakuake	kde.org/applications/system/org.kde.yakuake

Although many graphical terminal emulator applications are available, the focus in this chapter is on three. Installed in different Linux distributions by default, they are GNOME Terminal, Konsole Terminal, and xterm.

Using the GNOME Terminal Emulator

GNOME Terminal is the GNOME Shell desktop environment's default terminal emulator. Many distributions, such as Red Hat Enterprise Linux (RHEL), CentOS, and Ubuntu, use the GNOME Shell desktop environment by default, and therefore use GNOME Terminal by default. It is fairly easy to use and a good terminal emulator for individuals who are new to Linux. This section walks you through the various parts of accessing, configuring, and using GNOME Terminal.

Accessing GNOME Terminal

In the GNOME Shell desktop environment, accessing the GNOME Terminal is fairly straightforward. Click on the Activities icon in the upper-right corner of the desktop window. When the search bar appears, click within the bar to access it and type **terminal**. The results of these actions are shown in Figure 2-4.

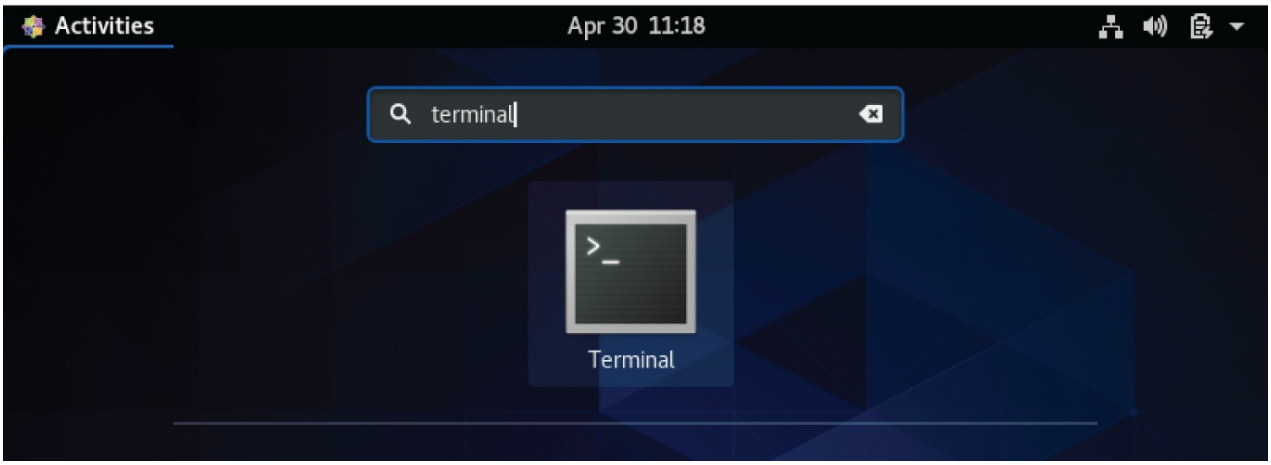

FIGURE 2-4 Finding GNOME Terminal in GNOME Shell

Notice in the previous figure that the GNOME Terminal application icon is named Terminal. Click the icon to open the terminal emulator. An open GNOME Terminal application on a CentOS distribution is shown in Figure 2-5.

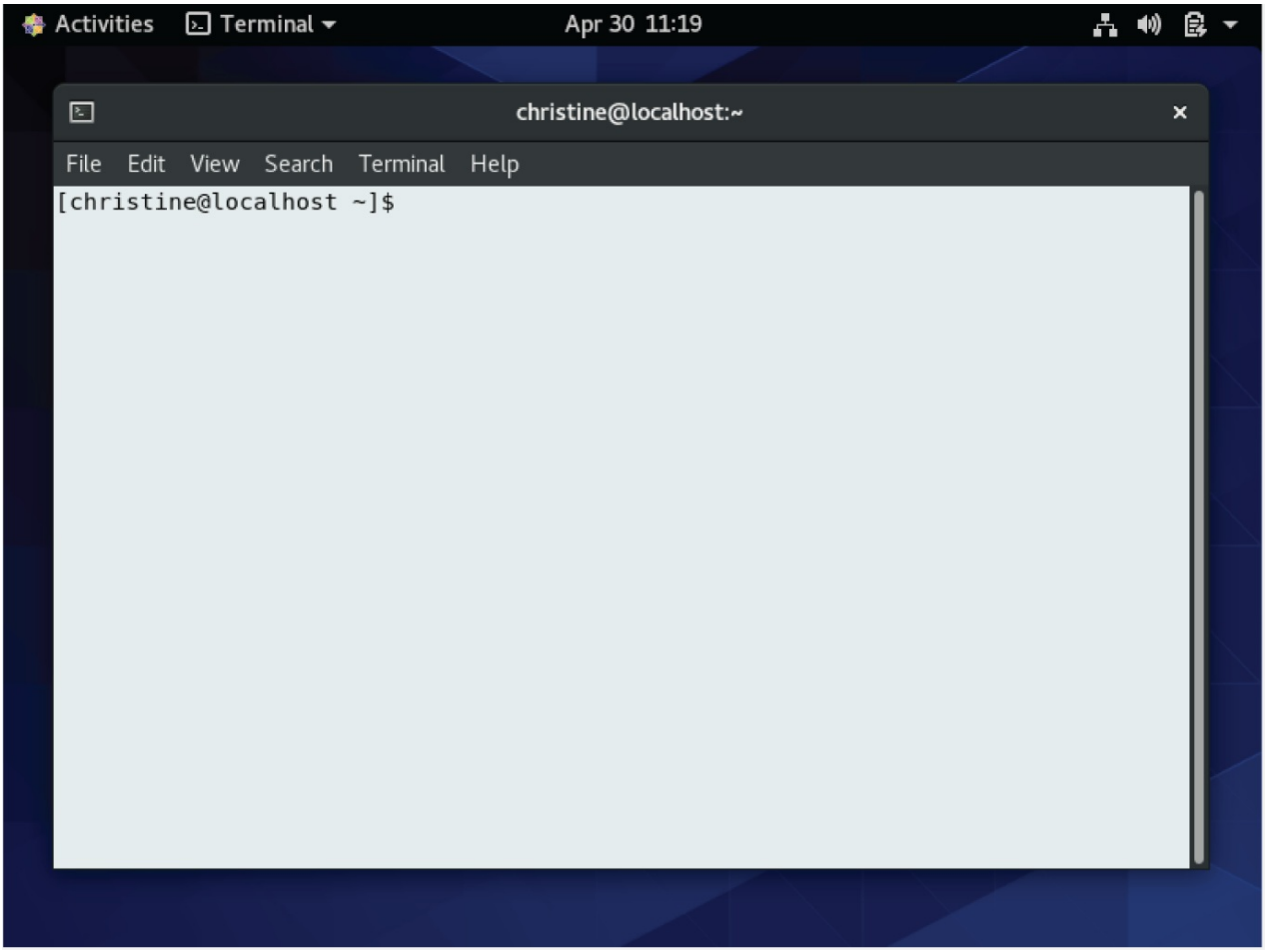

FIGURE 2-5 GNOME Terminal on CentOS

When you are done using the terminal emulator application, you close it just like other desktop windows: by clicking the x in the window's upper-right corner.

The GNOME Terminal application's appearance may vary between Linux distributions. For example, in Figure 2-6, GNOME Terminal is shown on an Ubuntu GNOME Shell desktop environment.

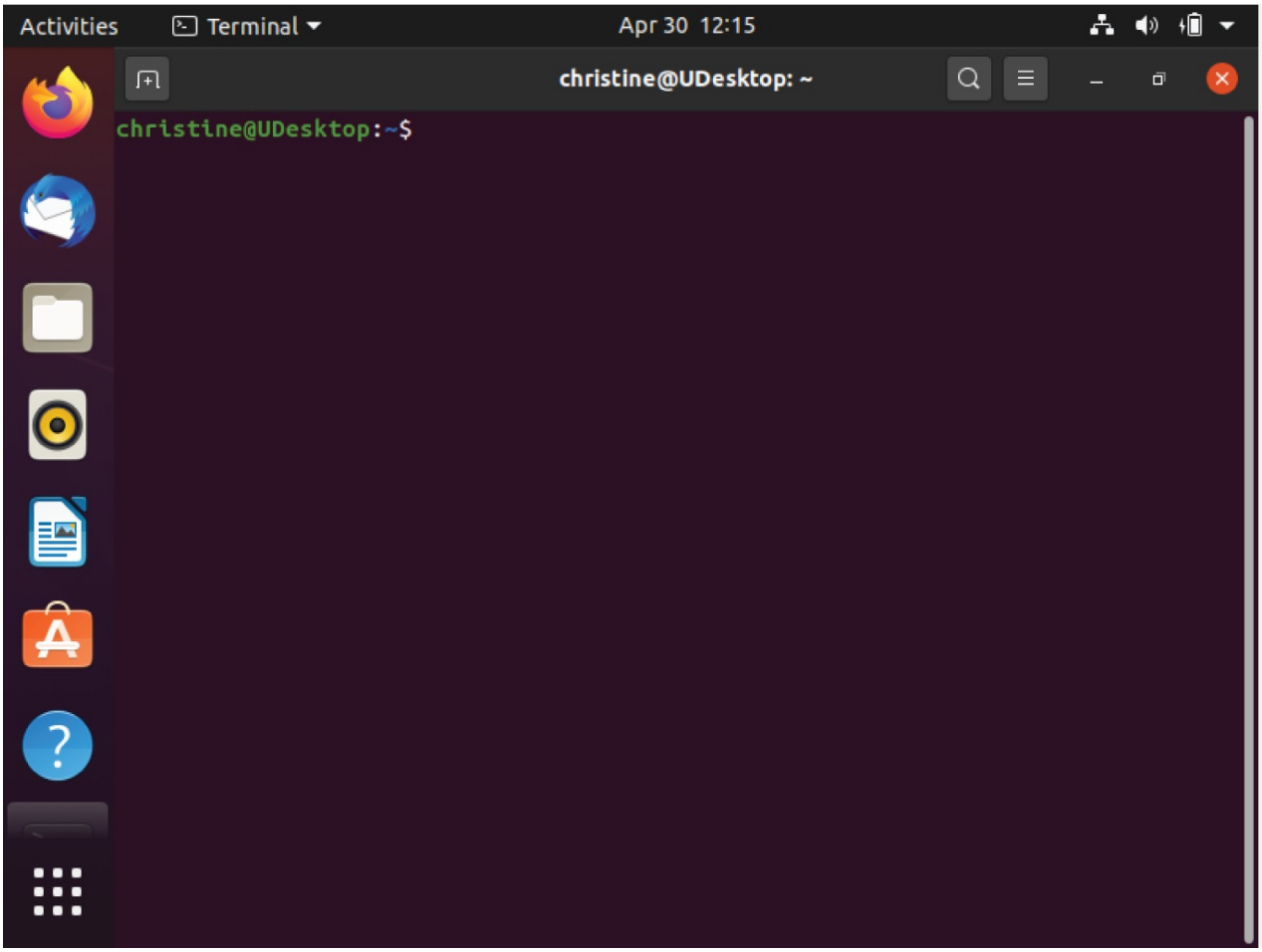

FIGURE 2-6 GNOME Terminal on Ubuntu

Notice that the appearance of the GNOME Terminal application in Figure 2-6 is different compared to Figure 2-5. This is typically due to the application's default configuration (covered later in this chapter) and the various features a Linux distribution has in its GUI windows.

> **TIP**
>
> **If you are using a different desktop environment than GNOME Shell (and have GNOME Terminal installed), be aware that you may not have a search feature. In these cases, use the environment's menu system to look for GNOME Terminal. It is typically named Terminal.**

On many distributions, the first time you launch the GNOME Terminal application, its terminal emulator icon will appear in your GNOME Shell Favorites bar. Hovering over the icon with your mouse will show the terminal emulator's name, as shown in Figure 2-7.

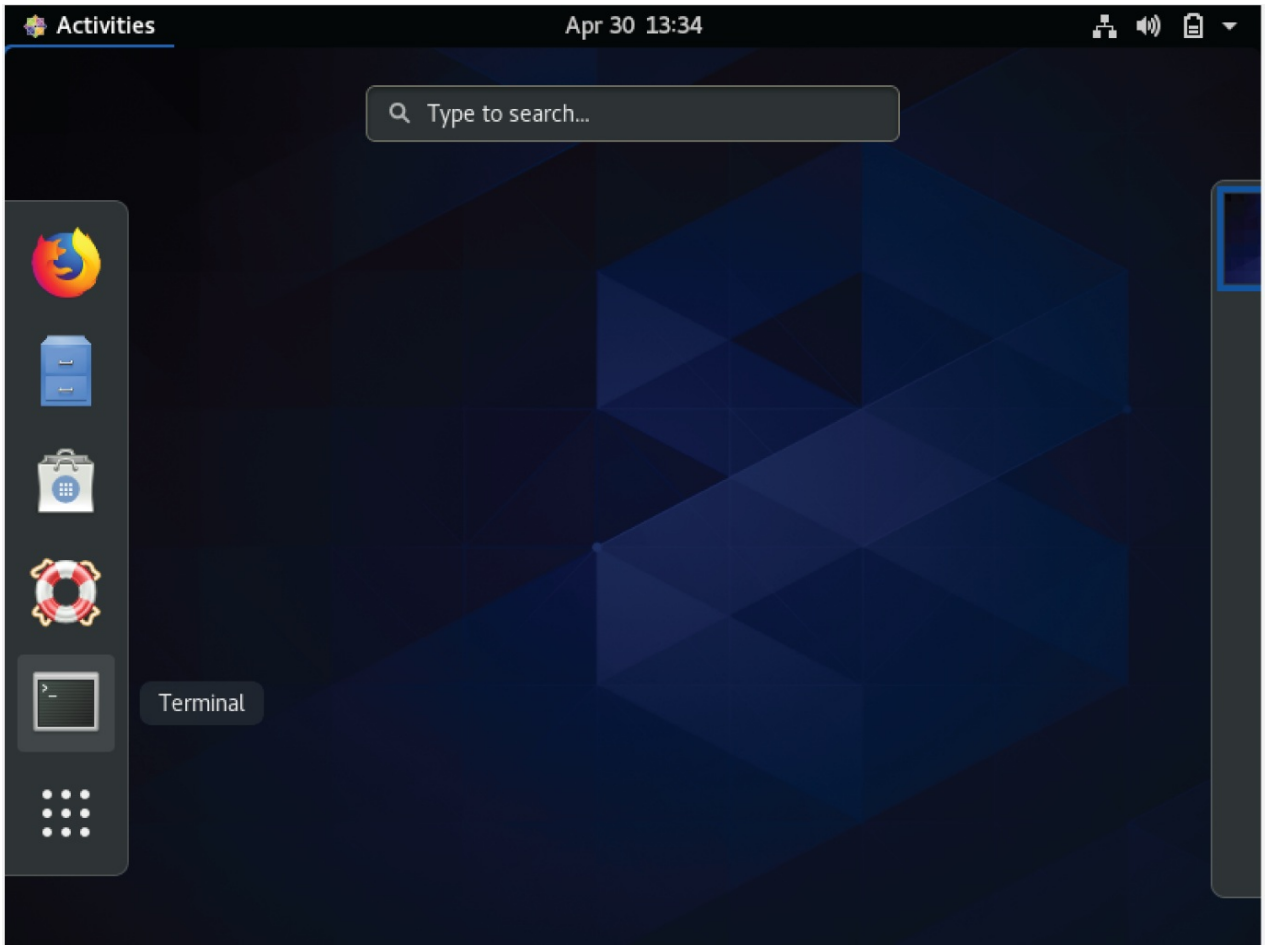

FIGURE 2-7 GNOME Terminal icon in the Favorites bar

If for some reason the icon does not show up in your Favorites bar, you can set up a keyboard shortcut to launch it. This approach is handy for those who don't care for using a mouse, and it allows faster access to the CLI.

> **TIP**
>
> **GNOME Shell on the Ubuntu distribution already has a keystroke shortcut for opening the GNOME terminal emulator: Ctrl+Alt+T.**

To create a keyboard shortcut, you'll need to access the Keyboard Shortcuts window within Keyboard Settings. To accomplish this quickly, click the Activities icon in the upper-right corner of the GNOME Shell desktop window. When the search bar appears, click within the bar to access it, and type **Keyboard Shortcuts**. The results of these actions are shown in Figure 2-8.

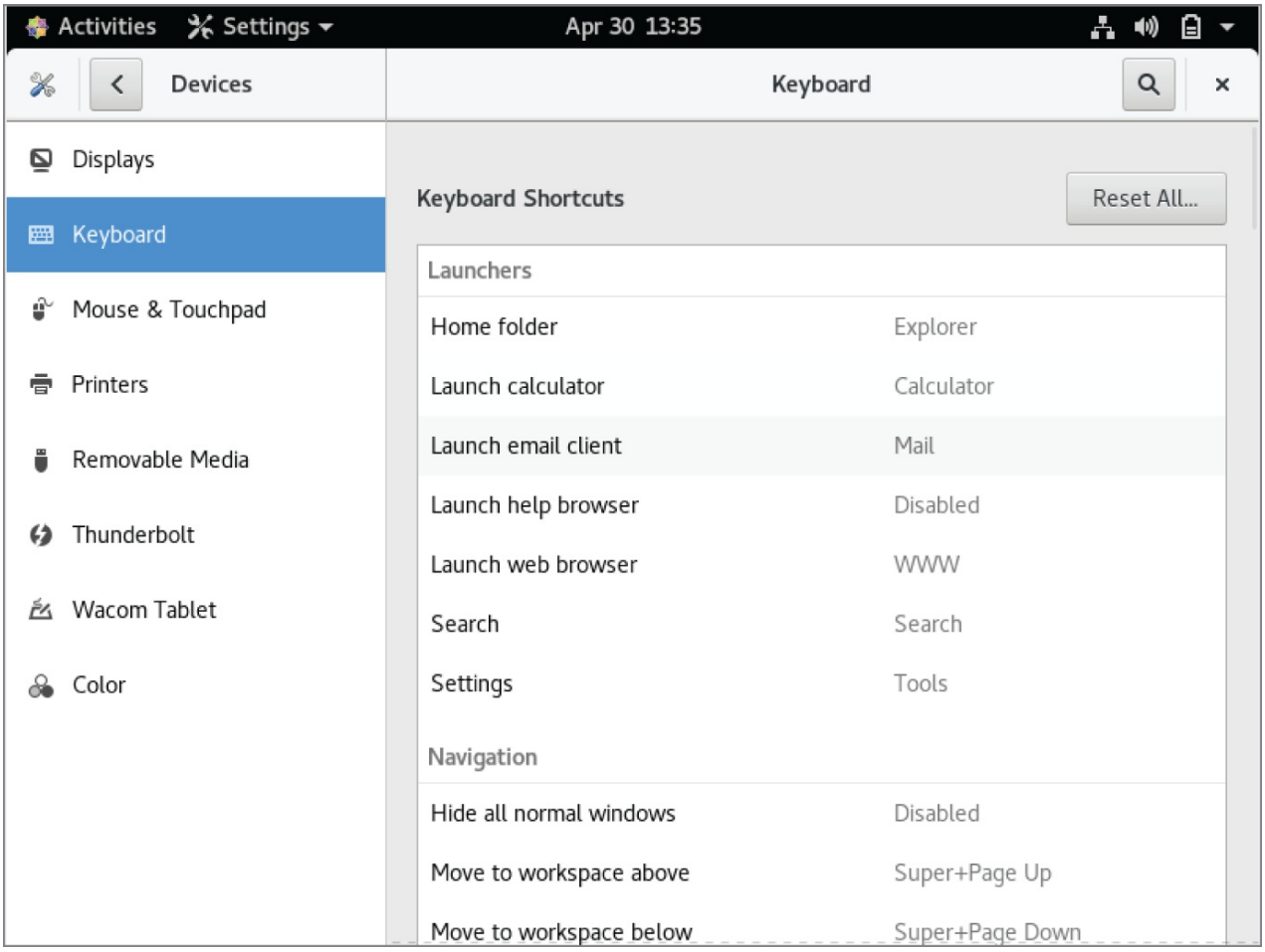

FIGURE 2-8 Reaching the Keyboard Shortcuts window

Once you are in the Keyboard Shortcuts window, scroll down to reach the + button, which is all the way at the bottom. Clicking this button opens a dialog box, where you can name your new shortcut, provide the command to open the application, and set the shortcut's keystrokes, as shown in Figure 2-9.

It is important to use the correct command name in order to properly launch the GNOME terminal emulator, so type **gnome-terminal** in the Command field, as shown in Figure 2-9. When you are all done setting up your new shortcut, click the Add button in the window. Now you can quickly launch the GNOME Terminal by just using the keystroke combination you specified.

Several configuration options are provided by menus and shortcut keys in the application, which you can apply after you get the GNOME terminal emulation started. Understanding these options can enhance your GNOME Terminal CLI experience.

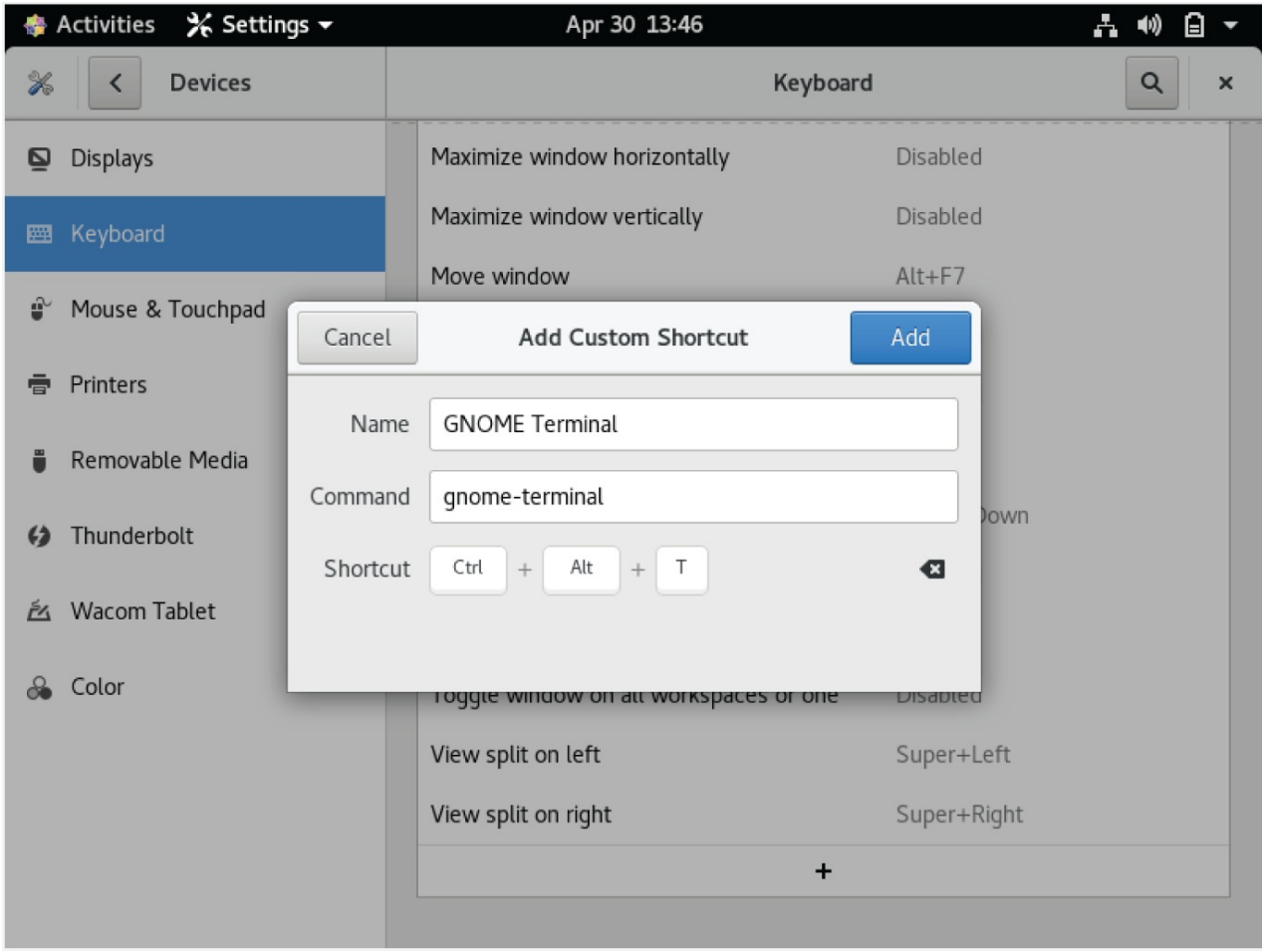

FIGURE 2-9 Creating a keyboard shortcut

The menu bar

The GNOME Terminal menu bar contains the configuration and customization options you need to make your GNOME Terminal just the way you want it. The following tables briefly describe the configuration options in the menu bar and shortcut keys associated with them.

> **TIP**
>
> **If the GNOME Terminal window does not display its menu bar, right-click in the terminal emulator session area, and click Show Menubar in the drop-down menu.**

Table 2-4 shows the configuration options available within the GNOME Terminal File menu system. The File menu item contains items to create and manage your overall CLI terminal sessions.

TABLE 2-4 The File Menu

Name	Shortcut Key	Description
New Tab	Shift+Ctrl+T	Starts a new shell session in a new tab in the existing GNOME Terminal window
New Window	Shift+Ctrl+N	Starts a new shell session in a new GNOME Terminal window
Close Tab	Shift+Ctrl+W	Closes the current tab in the GNOME Terminal window
Close Window	Shift+Ctrl+Q	Closes the current GNOME Terminal window

Notice that, as in a network browser, you can open new tabs within the GNOME Terminal session to start a whole new CLI session. Each tab session is considered to be an independent CLI session.

> **TIP**
>
> **You don't always have to click through the menu to reach options in the File menu. Some of the File menu selections are also available by right-clicking in the terminal emulator session area.**

The Edit menu contains items, shown in Table 2-5, for handling text within the tabs. You can copy and paste text

anywhere within the session window.

TABLE 2-5 The Edit Menu

Name	Shortcut Key	Description
Copy	Shift+Ctrl+C	Copies selected text to the GNOME clipboard
Copy as HTML	None	Copies selected text, along with its font and color, to the GNOME clipboard
Paste	Shift+Ctrl+V	Pastes text from the GNOME clipboard into a session
Select All	None	Selects output in the entire scrollback buffer
Preferences	None	Edits the current session profile

Copying and pasting commands in the terminal is useful if you are lacking in keyboarding skills. Thus, the keyboard shortcuts for the GNOME Terminal Copy and Paste functions are worth memorizing.

NOTE

As you read through these GNOME Terminal menu options, keep in mind that your Linux distribution's GNOME Terminal may have slightly different menu options available. This is because several Linux distributions use older versions of GNOME Terminal. You can find the version number by clicking Help in the menu bar and selecting About from the drop-down menu.

The View menu, shown in Table 2-6, contains items for controlling how the CLI session windows appear. These options can be helpful for individuals with visual impairment.

TABLE 2-6 The View Menu

Name	Shortcut Key	Description
Show Menubar	None	Toggles on/off the menu bar display
Full Screen	F11	Toggles on/off the terminal window filling the entire desktop
Zoom In	Ctrl++	Enlarges the font size in the window incrementally
Normal Size	Ctrl+0	Returns the font size to default
Zoom Out	Ctrl+-	Reduces the font size in the window incrementally

Be aware that if you toggle off the menu bar display, the session's menu bar disappears. However, you can easily get the menu bar to display again by right-clicking in any terminal session window and selecting the Show Menubar option.

The Search menu, shown in Table 2-7, contains items for conducting simple searches within the terminal session. These searches are similar to ones you may have conducted in a network browser or word processor.

TABLE 2-7 The Search Menu

Name	Shortcut Key	Description
Find	Shift+Ctrl+F	Opens Find window to provide designated text search options
Find Next	Shift+Ctrl+G	Searches forward from current terminal session location for designated text
Find Previous	Shift+Ctrl+H	Searches backward from current terminal session location for designated text
Clear Highlight	Shift+Ctrl+J	Removes highlighting of found text

The Terminal menu, shown in Table 2-8, contains options for controlling the terminal emulation session features. There are no shortcut keys to access these items.

TABLE 2-8 The Terminal Menu

Name	Description
Read-Only	Toggles on/off the terminal session accepting keyboard strokes; it does *not* enable/disable keyboard shortcuts
Reset	Sends reset terminal session control code
Reset and Clear	Sends reset terminal session control code and clears terminal session screen
80x24	Adjusts the current terminal window size to 80 columns wide by 24 rows high
80x43	Changes the current terminal window size to 80 columns wide by 43 rows high
132x24	Adjusts the current terminal window size to 130 columns wide by 24 rows high
130x43	Changes the current terminal window size to 130 columns wide by 43 rows high

The Reset option is extremely useful. One day, you may accidentally cause your terminal session to display random characters and symbols. When this occurs, the text is unreadable. It is typically caused by displaying a nontext file to the screen. You can quickly get the terminal session back to normal by selecting Reset or Reset And Clear.

> **NOTE**
>
> **Keep in mind that when you adjust your terminal's size, such as by using the 80x24 setting in the Terminal menu, the actual size is determined by factors such as the character font in use. It's a good idea to play around with the different settings to find a size that suits your taste.**

The Tabs menu, shown in Table 2-9, provides items for controlling the location of the tabs and selecting which tab is active. This menu displays *only* when you have more than one tab session open.

TABLE 2-9 The Tabs Menu

Name	Shortcut Key	Description
Previous Tab	Ctrl+Page Up	Makes the previous tab in the list active
Next Tab	Ctrl+Page Down	Makes the next tab in the list active
Move Terminal Left	Shift+Ctrl+Page Up	Shuffles the current tab in front of the previous tab
Move Terminal Right	Shift+Ctrl+Page Down	Shuffles the current tab in front of the next tab
Detach Terminal	None	Removes the tab and starts a new GNOME Terminal window using this tab session

Finally, the Help menu contains two menu options:

- Contents provides a full GNOME Terminal manual so that you can research individual GNOME Terminal items and features.
- About shows you the current GNOME Terminal application version that's running.

Besides the GNOME terminal emulator package, another commonly used package is Konsole. In many ways, Konsole is similar to GNOME Terminal. However, enough differences exist to warrant its own section.

Using the Konsole Terminal Emulator

The KDE project created its own terminal emulation package called *Konsole*. The Konsole application incorporates basic terminal emulation features, along with more advanced ones expected from a graphical application. This section describes Konsole features and shows you how to use them.

Accessing Konsole

The Konsole application is the default terminal emulator for the KDE desktop environment, Plasma. You can easily access it via the KDE environment's menu system. In other desktop environments, accessing Konsole is typically done via search features.

In the KDE desktop environment (Plasma), you start the Konsole terminal emulator by clicking the icon labeled Application Launcher in the lower-left corner of the screen. Then click Applications ➪ System ➪ Terminal (Konsole).

> **NOTE**
>
> **You may see two or more terminal menu options within the Plasma menu environment. If you do, the Terminal menu option with the word Konsole beneath it is the Konsole terminal emulator application.**

In the GNOME Shell desktop environment, the Konsole application is typically not installed by default. If Konsole has been installed, you can access it via the GNOME Shell search feature. Click the Activities icon in the upper-right corner of the desktop window. When the search bar appears, click your mouse within the bar to access it, and type **konsole**. If the terminal emulator is available on your system, you will see the Konsole icon displayed.

NOTE

You may not have the Konsole terminal emulation package installed on your system. If you would like to install it, see Chapter 9, "Installing Software," to learn how to install software via the command line.

Click the Konsole icon with your mouse to open the terminal emulator. An open Konsole application on an Ubuntu distribution is shown in Figure 2-10.

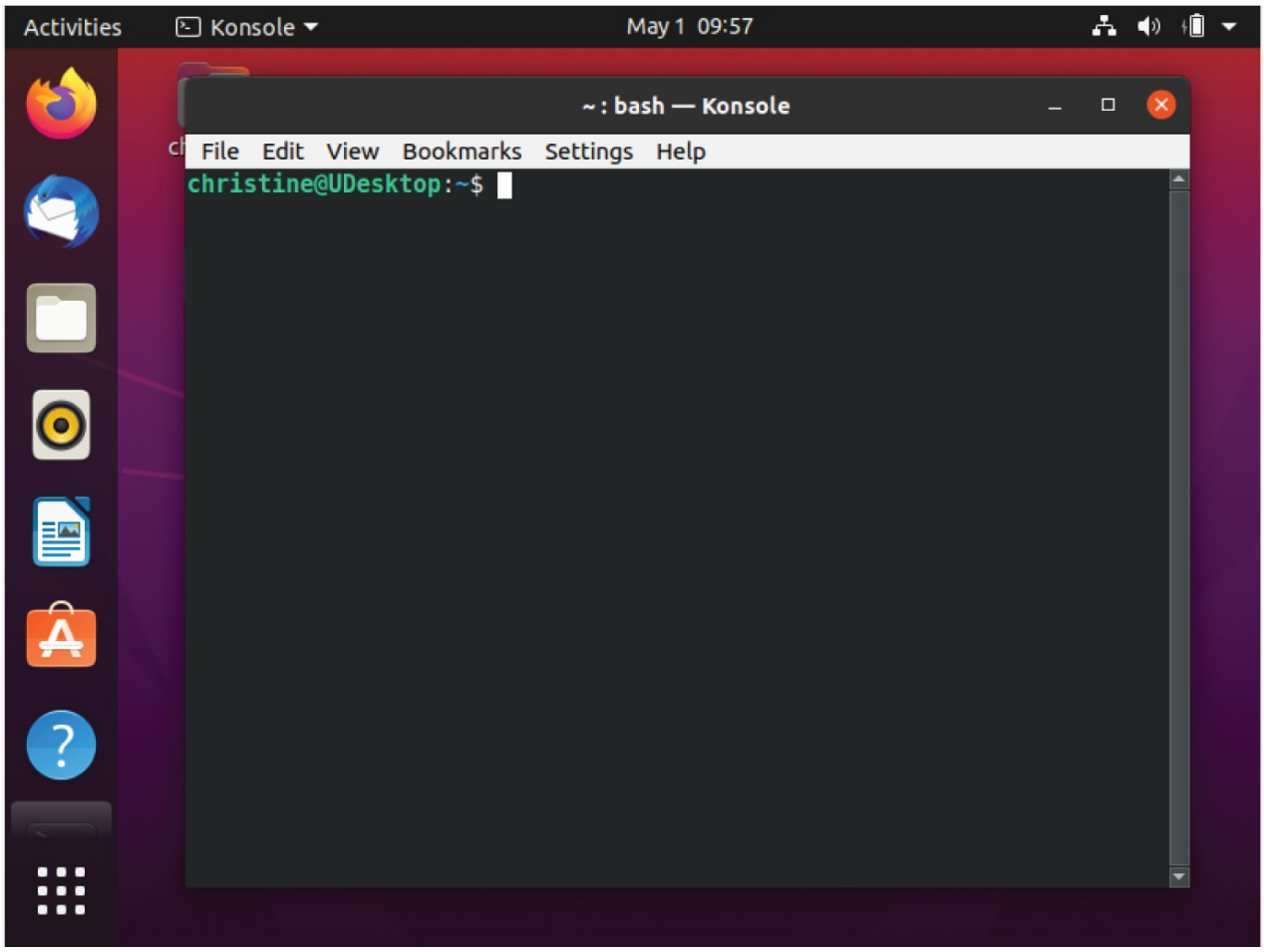

FIGURE 2-10 The Konsole terminal emulator

Remember that, in most desktop environments, you can create a keyboard shortcut to access applications such as Konsole. The command you need to type for the shortcut in order to start up the Konsole terminal emulator is **konsole**. Also, if the Konsole application is installed, you can start it from another terminal emulator by typing **konsole** and pressing Enter.

TIP

In the Plasma desktop environment, the Konsole terminal emulator application already has a default keyboard shortcut: Ctrl+Alt+T.

The Konsole terminal emulator, similar to GNOME Terminal, has several configuration options provided by menus and shortcut keys. The following section describes these various features.

The menu bar

The Konsole menu bar contains the configuration and customization options you need to easily view and change features in your terminal emulation session. The following tables briefly describe the menu options and associated shortcut keys.

TIP

If the Konsole menu bar is not currently displayed, you can press Ctrl+Shift+M to enable it.

The File menu, shown in Table 2-10, provides options for starting a new tab in the current window or in a new window.

TABLE 2-10 The File Menu

Name	Shortcut Key	Description
New Window	Ctrl+Shift+N	Starts a new shell session in a new Konsole Terminal window
New Tab	Ctrl+Shift+T	Starts a new shell session in a new tab in the existing Konsole Terminal window
Clone Tab	None	Starts a new shell session in a new tab in the existing Konsole Terminal window that attempts to duplicate the current tab
Save Output As	Ctrl+Shift+S	Saves the current tab's output in its scrollback buffer as either a text or an HTML file
Print Screen	Ctrl+Shift+P	Prints the current tab's displayed text
Open File Manager	None	Opens the default file browser application
Close Session	Ctrl+Shift+W	Closes the current tab session
Close Window	Ctrl+Shift+Q	Closes the current Konsole window

Notice that Konsole offers two handy options for saving information from your shell session: Save Output As and Print Screen. The Print Screen function allows you to print the displayed text to a system printer or save it as a PDF file.

NOTE

As you read through these Konsole menu options, keep in mind that your Linux distribution's Konsole application may have very different menu options available. This is because some Linux distributions have kept older versions of the Konsole terminal emulation package.

The Edit menu, shown in Table 2-11, provides options for handling text in the session. Also, managing tab names is in this options list.

TABLE 2-11 The Edit Menu

Name	Shortcut Key	Description
Copy	Ctrl+Shift+C	Copies selected text to the Konsole clipboard
Paste	Ctrl+Shift+V	Pastes text from the Konsole clipboard into a session
Select All	None	Selects all the text in the current tab
Copy Input To	None	Starts/stops session input copies to chosen additional sessions
Send Signal	None	Sends the selected signal from the drop-down menu to the current tab's shell process or other process
Rename Tab	Ctrl+Alt+S	Modifies session tab title bar setting
ZModem Upload	Ctrl+Alt+U	Starts the process of uploading a selected file, if the ZMODEM file transfer protocol is supported
Find	Ctrl+Shift+F	Opens the Find window to provide scrollback buffer text search options
Find Next	F3	Finds the next text match in more recent scrollback buffer history
Find Previous	Shift+F3	Finds the next text match in older scrollback buffer history

Konsole provides an excellent method for tracking what function is taking place in each tab session. Using the Rename Tab option, you can name a tab to match its current task. This helps in tracking which open tab session is performing what job.

NOTE

Konsole retains a history, formally called a *scrollback buffer*, for each tab. The history contains output text that has scrolled out of the terminal viewing area. By default, the last 1,000 lines in the scrollback buffer are retained. You can scroll back through the scrollback buffer by simply using the scrollbar in the viewing area. Also, you can scroll back line by line by pressing the Shift+Up Arrow or scroll back a page (24 lines) at a time by pressing Shift+Page Up.

The View menu, shown in Table 2-12, contains items for controlling individual session views in the Konsole Terminal window. In addition, options are available that aid in monitoring terminal session activity.

TABLE 2-12 The View Menu

Name	Shortcut Key	Description
Split View	None	Controls a multiple tab session display within the current Konsole window
Detach Current Tab	Ctrl+Shift+L	Removes a tab session and starts a new Konsole window using this tab session
Detach Current View	Ctrl+Shift+H	Removes the current tab session's view and starts a new Konsole window with it
Monitor for Silence	Ctrl+Shift+I	Toggles on/off a special message when no activity is occurring in the tab session
Monitor for Activity	Ctrl+Shift+A	Toggles on/off a special message when activity starts occurring in the tab session
Read-only	None	Toggles on/off the terminal session accepting keyboard strokes; does *not* enable/disable keyboard shortcuts
Enlarge Font	Ctrl++	Enlarges the font size in the window incrementally
Reset Font Size	Ctrl+Alt+0	Returns the font size to default
Shrink Font	Ctrl+-	Reduces the font size in the window incrementally
Set Encoding	None	Selects the character set used to send and display characters
Clear Scrollback	None	Removes the text in the current session's scrollback buffer
Clear Scrollback and Reset	Ctrl+Shift+K	Removes the text in the current session's scrollback buffer and resets the terminal window
Full Screen Mode	F11	Toggles on/off the terminal window filling the entire monitor display area

The Monitor for Silence option is used for indicating tab silence. Tab silence occurs when no new text appears in the current tab session for about seven seconds. This allows you to switch to another tab while waiting for the application's output to stop.

TIP

The Konsole application provides a simple menu when you right-click in the active session area. Several menu items are available in this easy-to-access menu.

The Bookmarks menu options, shown in Table 2-13, provide a way to manage *bookmarks* set in the Konsole window. A bookmark enables you to save your active session's directory location and then easily return there in either the same session or a new session.

TABLE 2-13 The Bookmarks Menu

Name	Shortcut Key	Description
Add Bookmark	Ctrl+Shift+B	Creates a new bookmark at the current directory location
Bookmark Tabs as Folder	None	Creates a new bookmark for all current terminal tab sessions
New Bookmark Folder	None	Creates a new bookmark storage folder
Edit Bookmarks	None	Edits existing bookmarks

The Settings menu, shown in Table 2-14, allows you to customize and manage your profiles. Profiles allow a user to automate the running of commands, set up the session's appearance, configure the scrollback buffer, and so on. Also, within the Settings menu you can add a little more functionality to your shell sessions.

TABLE 2-14 The Settings Menu

Name	Shortcut	Key Description
Edit Current Profile	None	Opens the Edit Profile window to provide profile configuration options
Switch Profile	None	Applies to the current tab a selected profile
Manage Profiles	None	Opens the Manage Profiles window to provide profile management options
Show Menubar	Ctrl+Shift+M	Toggles on/off menu bar display
Configure Keyboard Shortcuts	None	Creates Konsole command keyboard shortcuts
Configure Notifications	None	Creates custom Konsole notifications
Configure Konsole	Ctrl+Shift+,	Configures many Konsole features

Configure Notifications allows you to associate specific events that can occur within a session with different actions, such as playing a sound. When one of the events occurs, the defined action (or actions) is taken.

The Help menu, shown in Table 2-15, provides the full Konsole handbook (if KDE handbooks were installed in your Linux distribution) and the standard About Konsole dialog box.

TABLE 2-15 The Help Menu

Name	Shortcut Key	Description
Konsole Handbook	None	Contains the full Konsole Handbook
What's This?	Shift+F1	Contains help messages for terminal widgets
Report Bug	None	Opens the Submit Bug Report form
Donate	None	Opens the KDE donation page within a web browser
Switch Application Language	None	Opens the Switch Application Language form
About Konsole	None	Displays information about the Konsole application, including its current version
About KDE	None	Displays information about the KDE desktop environment

Rather extensive documentation is provided to help you use the Konsole terminal emulator package within the Help menu. The Bug Report form to submit to the Konsole developers when you encounter a program bug is handy.

The Konsole terminal emulator package is young compared to another popular package, xterm. In the next section, we explore the “old-timer” xterm.

Using the xterm Terminal Emulator

The oldest and most basic of terminal emulation packages is *xterm*. The xterm package has been around since before the original days of X Window, a historically popular display server, and it's still included by default in some distributions, such as openSUSE.

xterm is a full terminal emulation package, but it doesn't require many resources (such as memory) to operate. Because of this, the xterm package is still popular in Linux distributions designed to run on older hardware.

Although it doesn't offer many fancy features, the xterm package does one thing extremely well: it emulates older terminals, such as the Digital Equipment Corporation (DEC) VT102, VT220, and Tektronix 4014 terminals. For the VT102 and VT220 terminals, xterm can even emulate the VT series of color control codes, allowing you to use color in your scripts.

NOTE

The DEC VT102 and VT220 were dumb text terminals popular for connecting to Unix systems in the 1980s and early 1990s. A VT102/VT220 could display text and display rudimentary graphics using block mode graphics. This style of terminal access is still used in many business environments today, thus keeping VT102/VT220 emulation popular.

Figure 2-11 shows what the basic xterm display looks like running on a CentOS distribution's GNOME Shell environment, where it had to be manually installed. You can see that it is very basic.

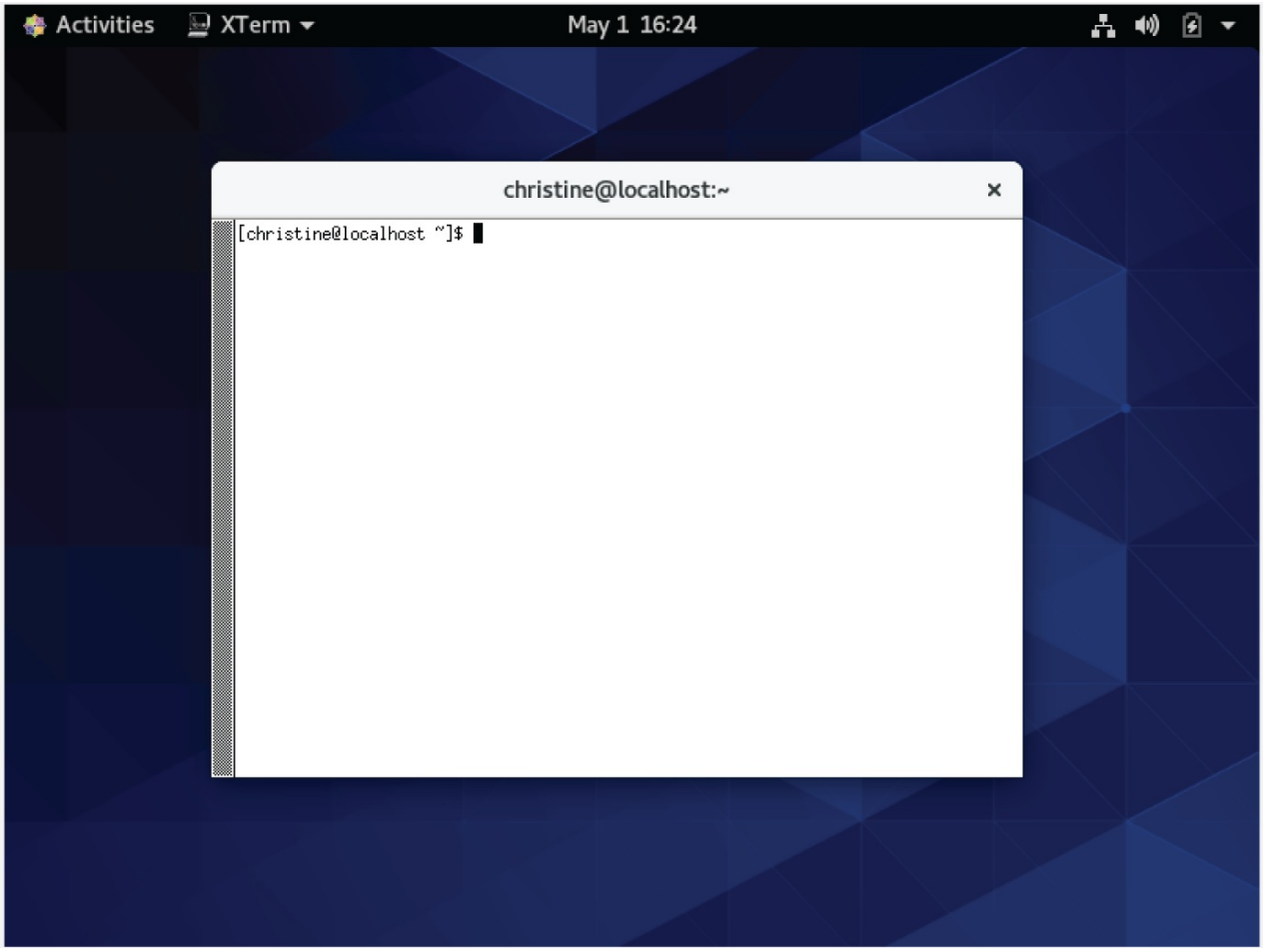

FIGURE 2-11 The xterm terminal

The xterm terminal emulator can be tricky to find these days. Often, it is not included in a desktop environment graphical menu arrangement.

Accessing xterm

In the KDE desktop environment (Plasma), you can access xterm by clicking the Application Launcher icon in the lower-left corner of the screen. Then click Applications ➪ System ➪ standard terminal emulator for the X Window system (xterm).

Once the xterm package is installed, you can access it via the GNOME Shell search feature. Click the Activities icon in the upper-right corner of the desktop window. When the search bar appears, click within the bar to access it and type **xterm**, and you'll see the Konsole icon displayed. Also, remember that you can create your own keyboard shortcut to start up xterm.

The xterm package allows you to set individual features using command-line parameters. The following sections discuss these features and how to change them.

Command-line parameters

The list of xterm command-line parameters is extensive. You can control lots of features to customize the terminal emulation features, such as enabling or disabling individual VT emulations.

> **NOTE**
>
> **xterm has a huge number of configuration options — so many that they cannot all be covered here. Extensive documentation is available via the Bash manual. Accessing the Bash manual is covered in Chapter 3. In addition, the xterm development team provides some excellent help on its website: `invisible-island.net/xterm`.**

You can invoke certain configuration options by adding a parameter to the `xterm` command. For example, to have the xterm emulate a DEC VT100 terminal, at the CLI type the command **`xterm -ti vt100`** and press Enter. Table 2-16 shows some parameters you can include when invoking the xterm terminal emulator software from the command line.

TABLE 2-16 xterm Command-Line Parameters

Parameter	Description
`-bg` *`color`*	Specifies the color to use for the terminal background
`-fb` *`font`*	Sets the font to use for bold text
`-fg` *`color`*	Specifies the color to use for the foreground text
`-fn` *`font`*	Sets the font to use for text
`-fw` *`font`*	Specifies the font to use for wide text
`-lf` *`filename`*	Sets the filename to use for screen logging
`-ms` *`color`*	Specifies the color used for the text cursor
`-` *`name`*	Sets the name of the application that appears in the title bar
`-ti` *`terminal`*	Specifies the terminal type to emulate

Some xterm command-line parameters use a plus sign (+) or minus sign (`-`) to signify how a feature is set. A plus sign may turn a feature on, whereas a minus sign turns it off. However, the opposite can be true as well. A plus sign may disable a feature, whereas a minus sign enables it, such as when using the `bc` parameter. Table 2-17 lists some of the more common features you can set using the `+` / `-` command-line parameters.

TABLE 2-17 xterm +/– Command-Line Parameters

Parameter	Description
`ah`	Enables/disables highlighted text cursor
`aw`	Enables/disables auto-line-wrap
`bc`	Enables/disables text cursor blinking
`cm`	Enables/disables recognition of ANSI color change control codes
`fullscreen`	Enables/disables full-screen mode
`j`	Enables/disables jump scrolling
`l`	Enables/disables logging screen data to a log file
`mb`	Enables/disables margin bell
`rv`	Enables/disables reverse video colors
`t`	Enables/disables Tektronix mode

It is important to note that not all implementations of xterm support all these command-line parameters. You can determine which parameters your xterm implements by using the `-help` parameter when you start xterm on your system.

NOTE

If xterm appeals to you but you'd like to use a more modern terminal emulation application, consider trying the rxvt-unicode package. It is available to install via most distributions' standard repositories (covered in Chapter 9), uses little RAM, and is very fast. Find out more at `software.schmorp.de/pkg/rxvt-unicode.html`.

Now that you have been introduced to three terminal emulator packages, the big question is, which is the best terminal emulator to use? There is no definite answer to that question. Which terminal emulator package you use depends on your individual needs and desires. But it is great to have so many choices.

Summary

To start learning Linux command-line commands, you need access to a CLI. In the world of graphical interfaces, this can sometimes be challenging. This chapter discussed various interfaces you should consider to get to the Linux command line.

First, this chapter discussed the difference between accessing the CLI via a virtual console terminal (a terminal outside the GUI) and a graphical terminal emulation package (a terminal inside the GUI). We took a brief look at the basic differences between these two access methods.

Next, we explored in detail accessing the CLI via a virtual console terminal, including specifics on how to change console terminal configuration options such as background color.

After looking at virtual console terminals, the chapter traveled through accessing the CLI via a graphical terminal emulator. Primarily, we covered three types of terminal emulators: GNOME Terminal, Konsole, and xterm.

This chapter also covered the GNOME Shell desktop project's GNOME terminal emulation package. GNOME

Terminal is typically installed by default on the GNOME Shell desktop environment. It provides convenient ways to set many terminal features through menu options and shortcut keys.

We also discussed the KDE desktop project's Konsole terminal emulation package. The Konsole application is typically installed by default on the KDE desktop environment (Plasma). It provides several nice features, such as the ability to monitor a terminal for silence.

Finally, we explored the xterm terminal emulator package. xterm was the first terminal emulator available for Linux. It can emulate older terminal hardware such as the VT and Tektronix terminals.

In the next chapter, we'll start looking at the Linux command-line commands. We'll walk you through the commands necessary to navigate around the Linux filesystem and to create, delete, and manipulate files.